

Statistical Modelling

Binary Logistic Regression

Complete Short Note: Theory, Equations and Interpretation

Purpose. This note reorganises all concepts in the Binary Logistic Regression lecture into a concise learning sequence. It explains what each equation means, how the inferential procedures are connected, and how model output should be interpreted.

Contents

1	Why Logistic Regression Is Needed	2
2	The Binary Logistic Regression Model	2
3	Odds, Log-Odds and the Logit Link	2
4	Estimating the Parameters	3
4.1	Bernoulli response	3
4.2	Log-likelihood and maximum likelihood	3
5	Inference for Individual Coefficients	4
5.1	Wald test	4
5.2	Confidence interval for a coefficient	4
6	Interpreting Coefficients Through Odds Ratios	4
6.1	Confidence interval for an odds ratio	5
7	Hosmer–Lemeshow Goodness-of-Fit Test	5
8	Likelihood-Ratio Test	5
9	Deviance and Nested-Model Comparison	6
9.1	Null and residual deviance	6
10	Pseudo-R^2 Measures	6
10.1	McFadden pseudo- R^2	7
10.2	Cox–Snell pseudo- R^2	7
10.3	Nagelkerke pseudo- R^2	7
11	From Probabilities to Predicted Classes	7
12	Compact Worked Example	8
13	Essential R Workflow	8
14	Exam-Focused Interpretation Checklist	9
15	Formula Summary	9

1 Why Logistic Regression Is Needed

Logistic regression models the relationship between one or more predictors and a *categorical* response. It estimates the probability that an observation belongs to a particular response category.

Examples include predicting whether:

- a manufactured component will develop a crack;
- a patient has a disease;
- a customer will purchase a product;
- a borrower will default on a loan.

The appropriate type depends on the response:

Type	Response structure	Example
Binary logistic	Exactly two categories	Disease: yes/no
Multinomial logistic	More than two unordered categories	Transport mode
Ordinal logistic	More than two ordered categories	Low/medium/high

This note focuses on **binary logistic regression**, where $Y_i \in \{0, 1\}$. Conventionally, $Y_i = 1$ is called the event of interest or success.

2 The Binary Logistic Regression Model

Let

$$\pi_i = P(Y_i = 1 \mid \mathbf{x}_i)$$

be the success probability for observation i . Define the linear predictor

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}.$$

The logistic model converts the unrestricted value $\eta_i \in (-\infty, \infty)$ into a valid probability:

$$\pi_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)} = \frac{1}{1 + \exp(-\eta_i)}$$

This inverse-logit expression is also called the **logistic** or **sigmoid** function. It guarantees $0 < \pi_i < 1$ because:

- $\exp(\eta_i) > 0$;
- the denominator is $1 + \exp(\eta_i)$, which is larger than the numerator.

With one predictor, the probability curve is theoretically S-shaped, approaching 0 and 1 asymptotically. A limited observed range of X may show only part of that curve.

Key point. The relationship is linear between the predictors and the *log-odds*, not between the predictors and the probability itself.

3 Odds, Log-Odds and the Logit Link

For an event with probability π , the odds are

$$\text{Odds} = \frac{\pi}{1 - \pi}.$$

For example, if $\pi = 0.80$, then

$$\text{Odds} = \frac{0.80}{0.20} = 4,$$

which may be written as 4 : 1.

Starting from the logistic probability model,

$$\frac{\pi_i}{1 - \pi_i} = \exp(\eta_i).$$

Taking natural logarithms gives

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}.$$

The transformation

$$g(\pi_i) = \log\left(\frac{\pi_i}{1 - \pi_i}\right)$$

is the **logit link**. Thus:

$$\text{probability} \xrightarrow{\text{logit}} \text{linear predictor}, \quad \text{linear predictor} \xrightarrow{\text{inverse logit}} \text{probability}.$$

4 Estimating the Parameters

4.1 Bernoulli response

For a binary observation,

$$Y_i \sim \text{Bernoulli}(\pi_i), \quad P(Y_i = y_i) = \pi_i^{y_i} (1 - \pi_i)^{1-y_i}.$$

For n independent observations, the likelihood is

$$L(\beta) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i}.$$

Substituting

$$\pi_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}, \quad 1 - \pi_i = \frac{1}{1 + \exp(\eta_i)}$$

gives

$$L(\beta) = \prod_{i=1}^n \left[\frac{\exp(\eta_i)}{1 + \exp(\eta_i)} \right]^{y_i} \left[\frac{1}{1 + \exp(\eta_i)} \right]^{1-y_i}.$$

4.2 Log-likelihood and maximum likelihood

The log-likelihood is

$$\ell(\beta) = \sum_{i=1}^n [y_i \log(\pi_i) + (1 - y_i) \log(1 - \pi_i)].$$

After substituting the logistic model:

$$\ell(\beta) = \sum_{i=1}^n [y_i \eta_i - \log\{1 + \exp(\eta_i)\}].$$

The maximum-likelihood estimate is

$$\hat{\beta} = \arg \max_{\beta} \ell(\beta).$$

Unlike ordinary least squares, logistic-regression estimates generally do not have a simple closed-form solution. Software finds them using an iterative numerical method.

Key point. A fitted model has a log-likelihood at least as large as that of a nested reduced model. Log-likelihoods are often negative, so larger means less negative.

5 Inference for Individual Coefficients

5.1 Wald test

For predictor X_j , the usual hypotheses are

$$H_0 : \beta_j = 0 \quad \text{versus} \quad H_1 : \beta_j \neq 0.$$

The Wald statistic is

$$z = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}.$$

Under H_0 , z is approximately standard normal. For a two-sided test,

$$p = 2P(Z \geq |z|).$$

Reject H_0 at level α when $p < \alpha$.

5.2 Confidence interval for a coefficient

An approximate $100(1 - \alpha)\%$ confidence interval is

$$\hat{\beta}_j \pm z_{\alpha/2} SE(\hat{\beta}_j).$$

For a 95% interval, $z_{\alpha/2} \approx 1.96$.

6 Interpreting Coefficients Through Odds Ratios

Holding all other predictors constant, increasing X_j by one unit changes the log-odds by β_j . Exponentiating gives the odds ratio:

$$OR_j = \exp(\beta_j).$$

Odds ratio	Interpretation
$OR_j > 1$	The odds increase as X_j increases.
$OR_j < 1$	The odds decrease as X_j increases.
$OR_j = 1$	A one-unit change produces no change in the odds.

The percentage change in odds is:

$$\begin{cases} (OR_j - 1) \times 100\%, & OR_j > 1, \\ (1 - OR_j) \times 100\% \text{ decrease}, & OR_j < 1. \end{cases}$$

Continuous predictor example. If $OR_{\text{Age}} = 1.08$, each additional year multiplies the odds by 1.08, an approximate 8% increase, holding other predictors constant.

Categorical predictor example. If Female is the reference category and $OR_{\text{Male}} = 1.50$, the odds for males are 1.5 times those for females, holding the other predictors constant.

6.1 Confidence interval for an odds ratio

If the confidence interval for β_j is

$$\hat{\beta}_j \pm z_{\alpha/2} SE(\hat{\beta}_j),$$

then exponentiate both limits:

$$\left[\exp\left\{\hat{\beta}_j - z_{\alpha/2} SE(\hat{\beta}_j)\right\}, \exp\left\{\hat{\beta}_j + z_{\alpha/2} SE(\hat{\beta}_j)\right\} \right].$$

If a confidence interval for an odds ratio contains 1, the corresponding two-sided coefficient test will generally not be significant at the matching level.

7 Hosmer–Lemeshow Goodness-of-Fit Test

The Hosmer–Lemeshow test examines whether predicted probabilities agree adequately with observed outcomes:

$$H_0 : \text{the model fits the data adequately}, \quad H_1 : \text{the model does not fit adequately}.$$

Observations are ordered by predicted probability and commonly divided into $g = 10$ groups. Because probabilities can be tied, the groups are not always exactly equal in size. Within each group, observed and expected event counts are compared.

A common form of the statistic is

$$\hat{C} = \sum_{k=1}^g \frac{(O_k - E_k)^2}{n_k \bar{\pi}_k (1 - \bar{\pi}_k)},$$

where O_k is the observed number of events, $E_k = n_k \bar{\pi}_k$ is the expected number, and n_k is the group size. Conventionally,

$$\hat{C} \sim \chi_{g-2}^2.$$

- Small p -value: evidence of poor fit; reject H_0 .
- Large p -value: insufficient evidence of poor fit; do not reject H_0 .

Key point. A large Hosmer–Lemeshow p -value does not prove that the model is perfect. It only means that this test did not detect a significant lack of fit.

8 Likelihood-Ratio Test

The likelihood-ratio test (LRT) compares two **nested** models:

- the reduced model contains fewer predictors;
- the full model contains all reduced-model predictors plus additional terms.

Let ℓ_R and ℓ_F be the maximised log-likelihoods. The test statistic is

$$G^2 = -2 \log\left(\frac{L_R}{L_F}\right) = -2(\ell_R - \ell_F) = 2(\ell_F - \ell_R).$$

Under H_0 ,

$$G^2 \sim \chi_{df}^2, \quad df = k_F - k_R,$$

where k_F and k_R are the numbers of estimated parameters.

H_0 : the additional predictors do not improve the model,

H_1 : the additional predictors improve the model.

	Wald test	Likelihood-ratio test
Main question	Is one estimated coefficient different from zero?	Does adding one or more parameters improve model fit?
Information used	Estimate and its standard error	Maximised likelihoods of two nested models
Advantage	Simple; printed in standard model output	Natural for testing several coefficients together

The LRT may behave better than the Wald test when estimates are very large or the normal approximation to an estimate is weak.

9 Deviance and Nested-Model Comparison

Deviance measures how far a fitted model is from the saturated model:

$$D = -2 [\ell(\text{fitted}) - \ell(\text{saturated})].$$

The saturated model reproduces the observed responses as closely as possible. A smaller deviance therefore indicates a model closer to the saturated model.

9.1 Null and residual deviance

- **Null deviance:** deviance of the intercept-only model.
- **Residual deviance:** deviance of the model containing the selected predictors.

The reduction is

$$\text{Null deviance} - \text{Residual deviance}.$$

A large reduction suggests that the predictors improve fit relative to the intercept-only model.

For reduced and full nested models:

$$\Delta D = D_R - D_F.$$

Because deviance is based on log-likelihood,

$$\Delta D = 2(\ell_F - \ell_R) = G^2.$$

Thus,

$$\Delta D \sim \chi^2_{k_F - k_R}.$$

Reject the null hypothesis when

$$p = P(\chi^2_{df} \geq \Delta D) < \alpha.$$

Key point. Likelihood ratio, difference in deviance and the chi-square comparison are three descriptions of the same nested-model test.

10 Pseudo- R^2 Measures

Ordinary linear-regression R^2 is based on sums of squares and the proportion of variation explained in a continuous response. Logistic regression uses a binary response and maximum likelihood, so ordinary R^2 does not directly apply.

10.1 McFadden pseudo- R^2

$$R_{\text{McFadden}}^2 = 1 - \frac{\ell_F}{\ell_0}$$

where $\ell_F = \log(L_F)$ is the fitted/full-model log-likelihood and $\ell_0 = \log(L_0)$ is the null-model log-likelihood.

Important correction. The lecture slide displays the ratio in the reverse order. The standard McFadden formula uses ℓ_F/ℓ_0 , not ℓ_0/ℓ_F . Since $\ell_F \geq \ell_0$ and both are commonly negative, the standard formula gives a non-negative improvement measure.

A value near zero indicates little improvement over the null model. Larger values indicate greater relative improvement, but McFadden R^2 should not be interpreted as a percentage of variance explained.

10.2 Cox–Snell pseudo- R^2

$$R_{\text{CS}}^2 = 1 - \left(\frac{L_0}{L_F} \right)^{2/n} = 1 - \exp \left[\frac{2}{n} (\ell_0 - \ell_F) \right].$$

It measures improvement over the null model, but its maximum is generally less than 1 for binary outcomes.

10.3 Nagelkerke pseudo- R^2

Nagelkerke rescales Cox–Snell so its theoretical maximum is 1:

$$R_N^2 = \frac{R_{\text{CS}}^2}{1 - L_0^{2/n}}.$$

It is often easier to communicate because its range extends from 0 to 1. Nevertheless, it remains a model-comparison measure, not ordinary variance explained.

Measure	Main idea	Important limitation
McFadden	Relative log-likelihood improvement	Not ordinary R^2
Cox–Snell	Likelihood improvement	Cannot generally reach 1
Nagelkerke	Rescaled Cox–Snell	Still not variance explained

11 From Probabilities to Predicted Classes

For observation i , logistic regression first produces

$$\hat{\pi}_i = P(Y_i = 1 \mid \mathbf{x}_i).$$

A classification threshold c converts this probability into a class:

$$\hat{Y}_i = \begin{cases} 1, & \hat{\pi}_i \geq c, \\ 0, & \hat{\pi}_i < c. \end{cases}$$

Although $c = 0.50$ is common, it is not automatically optimal. For example, if a disease probability is 0.35, a threshold of 0.50 classifies the patient as negative. If missing a true disease case is costly, $c = 0.30$ may be more appropriate.

Change	Likely benefit	Likely cost
Lower c	More positive cases detected	More false positives
Raise c	Fewer false positives	More false negatives

Threshold selection should therefore reflect the objective of the model and the relative consequences of false positives and false negatives.

12 Compact Worked Example

Suppose a disease model is

$$\text{logit}(\pi) = -4 + 0.06(\text{Age}) + 0.80(\text{RiskGroup}),$$

where RiskGroup = 1 for the exposed group and 0 for the reference group.

For a 50-year-old exposed person:

$$\eta = -4 + 0.06(50) + 0.80 = -0.20,$$

$$\hat{\pi} = \frac{\exp(-0.20)}{1 + \exp(-0.20)} \approx 0.450.$$

Interpretations:

- $OR_{\text{Age}} = \exp(0.06) \approx 1.062$: each additional year increases the odds by about 6.2%, holding risk group constant.
- $OR_{\text{RiskGroup}} = \exp(0.80) \approx 2.23$: the exposed group's odds are about 2.23 times those of the reference group at the same age.
- With $c = 0.50$, the person is classified as 0; with $c = 0.40$, the person is classified as 1.

13 Essential R Workflow

```
# Fit the full and null models
full_model <- glm(Y ~ X1 + X2, data = dat,
                  family = binomial(link = "logit"))
null_model <- glm(Y ~ 1, data = dat,
                  family = binomial(link = "logit"))

# Coefficients and Wald tests
summary(full_model)

# Coefficient confidence intervals and odds ratios
confint(full_model)
exp(coef(full_model))
exp(confint(full_model))

# Predicted probabilities and classes
prob <- predict(full_model, type = "response")
pred_class <- ifelse(prob >= 0.50, 1, 0)
table(Observed = dat$Y, Predicted = pred_class)

# Likelihood-ratio / deviance comparison
anova(null_model, full_model, test = "Chisq")

# Hosmer-Lemeshow test
ResourceSelection::hoslem.test(dat$Y, fitted(full_model))
```

14 Exam-Focused Interpretation Checklist

When interpreting binary logistic regression output, answer these questions in order:

1. What outcome is coded as $Y = 1$?
2. Is each coefficient positive or negative?
3. Is the Wald p -value below the chosen significance level?
4. What is the odds ratio $\exp(\hat{\beta}_j)$?
5. Does its confidence interval contain 1?
6. Does the full model improve on the reduced or null model according to the LRT/deviance test?
7. Does the Hosmer–Lemeshow test detect lack of fit?
8. Which pseudo- R^2 measure is being reported?
9. What classification threshold is used, and is it suitable for the practical cost of errors?

15 Formula Summary

$$\begin{aligned}\eta_i &= \beta_0 + \sum_{j=1}^p \beta_j x_{ij}, \\ \pi_i &= \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}, \\ \text{logit}(\pi_i) &= \log\left(\frac{\pi_i}{1 - \pi_i}\right) = \eta_i, \\ L(\boldsymbol{\beta}) &= \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i}, \\ \ell(\boldsymbol{\beta}) &= \sum_{i=1}^n [y_i \eta_i - \log\{1 + \exp(\eta_i)\}], \\ z &= \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}, \\ OR_j &= \exp(\hat{\beta}_j), \\ G^2 &= 2(\ell_F - \ell_R) \sim \chi_{k_F - k_R}^2, \\ \Delta D &= D_R - D_F = G^2, \\ R_{\text{McFadden}}^2 &= 1 - \frac{\ell_F}{\ell_0}, \\ R_{\text{CS}}^2 &= 1 - \left(\frac{L_0}{L_F}\right)^{2/n}, \\ R_{\text{N}}^2 &= \frac{R_{\text{CS}}^2}{1 - L_0^{2/n}}.\end{aligned}$$

End of Note